A1.3 AXI Architecture

The AXI protocol is burst-based and defines the following independent transaction channels:

- read address
- read data
- write address
- write data
- write response.

An address channel carries control information that describes the nature of the data to be transferred. The data is transferred between master and slave using either:

- A write data channel to transfer data from the master to the slave. In a write transaction, the slave uses the write response channel to signal the completion of the transfer to the master.
- A read data channel to transfer data from the slave to the master.

The AXI protocol:

- permits address information to be issued ahead of the actual data transfer
- supports multiple outstanding transactions
- supports out-of-order completion of transactions.

Figure A1-1 shows how a read transaction uses the read address and read data channels.

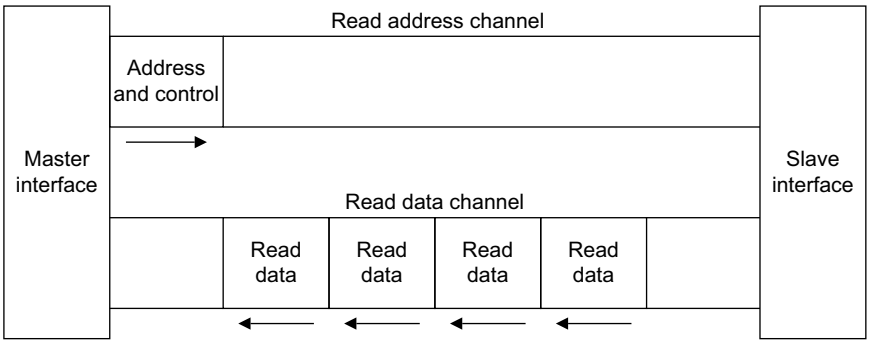


Figure A1-1 Channel architecture of reads

Figure A1-2 shows how a write transaction uses the write address, write data, and write response channels.

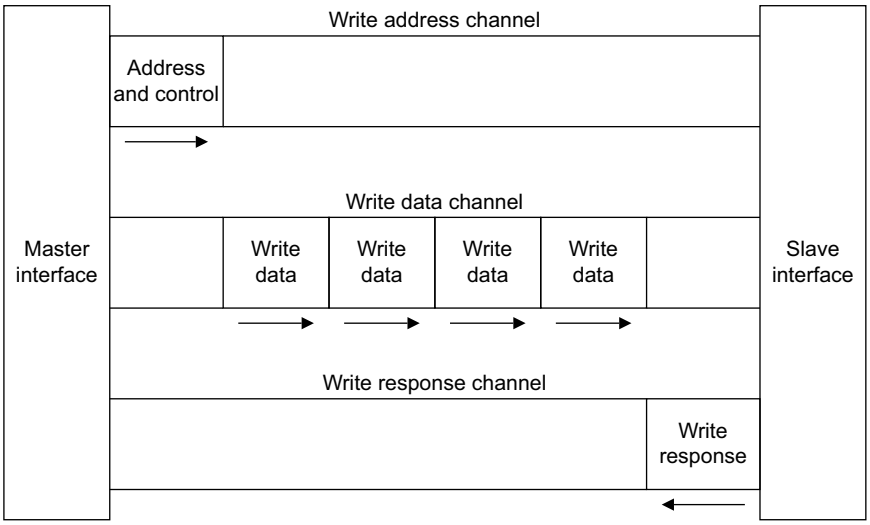


Figure A1-2 Channel architecture of writes

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A1.3.1 Channel definition

Each of the independent channels consists of a set of information signals and **VALID** and **READY** signals that provide a two-way handshake mechanism. See *Basic read and write transactions* on page A3-37.

The information source uses the **VALID** signal to show when valid address, data or control information is available on the channel. The destination uses the **READY** signal to show when it can accept the information. Both the read data channel and the write data channel also include a **LAST** signal to indicate the transfer of the final data item in a transaction.

Read and write address channels

Read and write transactions each have their own address channel. The appropriate address channel carries all of the required address and control information for a transaction.

Read data channel

The read data channel carries both the read data and the read response information from the slave to the master, and includes:

- the data bus, that can be 8, 16, 32, 64, 128, 256, 512, or 1024 bits wide
- a read response signal indicating the completion status of the read transaction.

Write data channel

The write data channel carries the write data from the master to the slave and includes:

- the data bus, that can be 8, 16, 32, 64, 128, 256, 512, or 1024 bits wide
- a byte lane strobe signal for every eight data bits, indicating which bytes of the data are valid.

Write data channel information is always treated as buffered, so that the master can perform write transactions without slave acknowledgement of previous write transactions.

Write response channel

A slave uses the write response channel to respond to write transactions. All write transactions require completion signaling on the write response channel.

As Figure A1-2 on page A1-22 shows, completion is signaled only for a complete transaction, not for each data transfer in a transaction.

A1.3.2 Interface and interconnect

A typical system consists of a number of master and slave devices connected together through some form of interconnect, as Figure A1-3 shows.

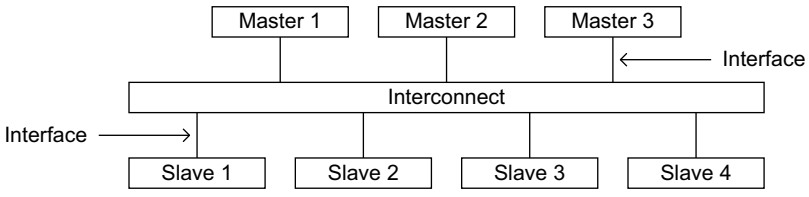


Figure A1-3 Interface and interconnect

The AXI protocol provides a single interface definition, for the interfaces:

- between a master and the interconnect
- between a slave and the interconnect
- between a master and a slave.

This interface definition supports a variety of different interconnect implementations.

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Note

An interconnect between devices is equivalent to another device with symmetrical master and slave ports to which real master and slave devices can be connected.

Typical system topologies

Most systems use one of three interconnect topologies:

- shared address and data buses
- shared address buses and multiple data buses
- multilayer, with multiple address and data buses.

In most systems, the address channel bandwidth requirement is significantly less than the data channel bandwidth requirement. Such systems can achieve a good balance between system performance and interconnect complexity by using a shared address bus with multiple data buses to enable parallel data transfers.

A1.3.3 Register slices

Each AXI channel transfers information in only one direction, and the architecture does not require any fixed relationship between the channels. This means a register slice can be inserted at almost any point in any channel, at the cost of an additional cycle of latency.

Note

This makes possible:

- a trade-off between cycles of latency and maximum frequency of operation
- a direct, fast connection between a processor and high performance memory, but to use simple register slices to isolate a longer path to less performance critical peripherals.

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